Análise dos Dados de Diabetes

```
library(ggplot2)
library(dplyr)
library(tidyr)
library(hnp)
library(pROC)
library(knitr)
library(here)
```

```
df = read.csv(here("dados", "diabetes.csv"))
df$Outcome = as.factor(df$Outcome)
df$classe = factor(df$Outcome,
                     levels = c(0,1),
                     labels = c("Não diabético", "Diabético"))

kable(head(df), caption = "Dados sobre Diabetes")
```

Table 1: Dados sobre Diabetes

Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome	classe
6	148	72	35	0	33.6	0.627	50	1	Diabético
1	85	66	29	0	26.6	0.351	31	0	Não diabético
8	183	64	0	0	23.3	0.672	32	1	Diabético
1	89	66	23	94	28.1	0.167	21	0	Não diabético
0	137	40	35	168	43.1	2.288	33	1	Diabético
5	116	74	0	0	25.6	0.201	30	0	Não diabético

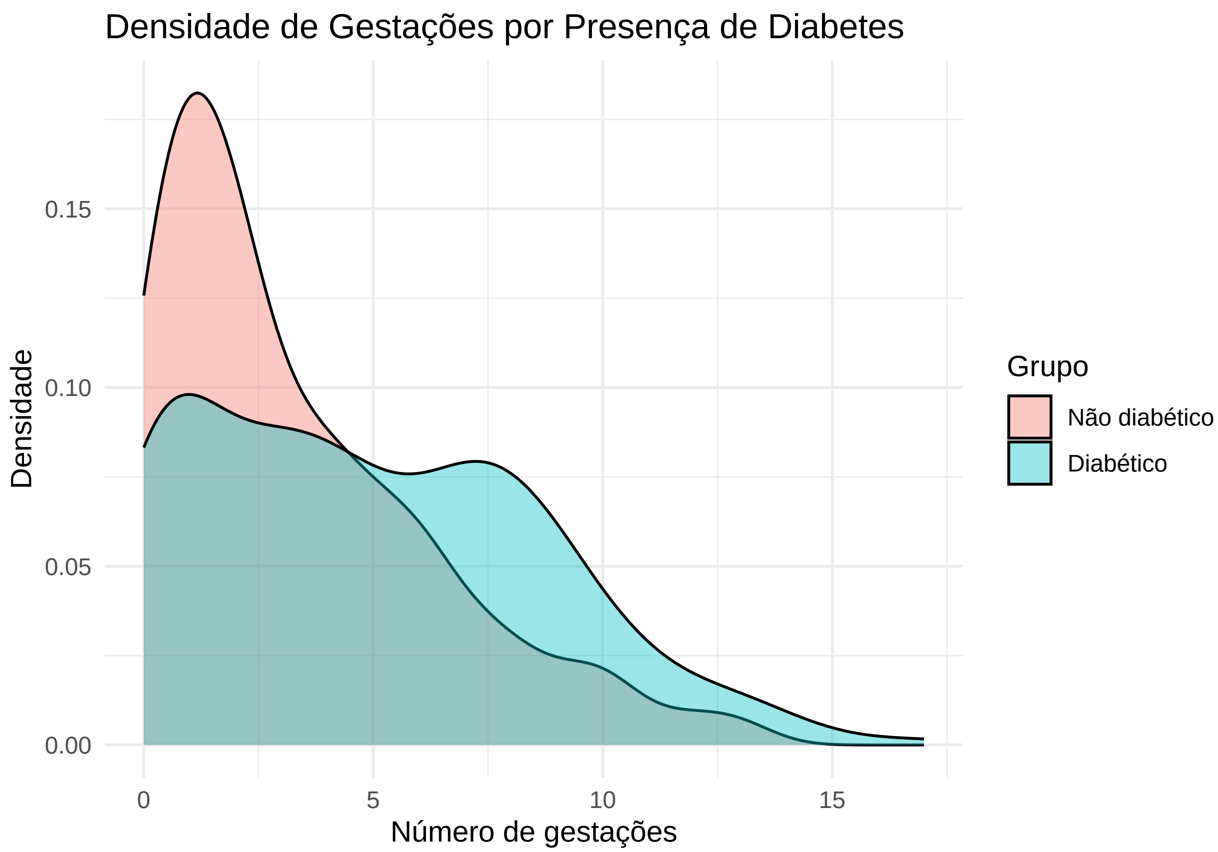
Pregnancies

```
descritivas = c("Outcome", "Média","DP","Med","Min","Máx","CoefVar")

df %>%
  group_by(Outcome) %>%
  summarise(
    Media = mean(Pregnancies, na.rm = TRUE),
    DP = sd(Pregnancies, na.rm = TRUE),
    Med = median(Pregnancies, na.rm = TRUE),
    Min = min(Pregnancies, na.rm = TRUE),
    Max = max(Pregnancies, na.rm = TRUE)
  ) %>%
  mutate(CoefVar = DP / Media * 100) %>%
  kable(caption = "Medidas Descritivas para Gestações",
        col.names = descritivas)
```

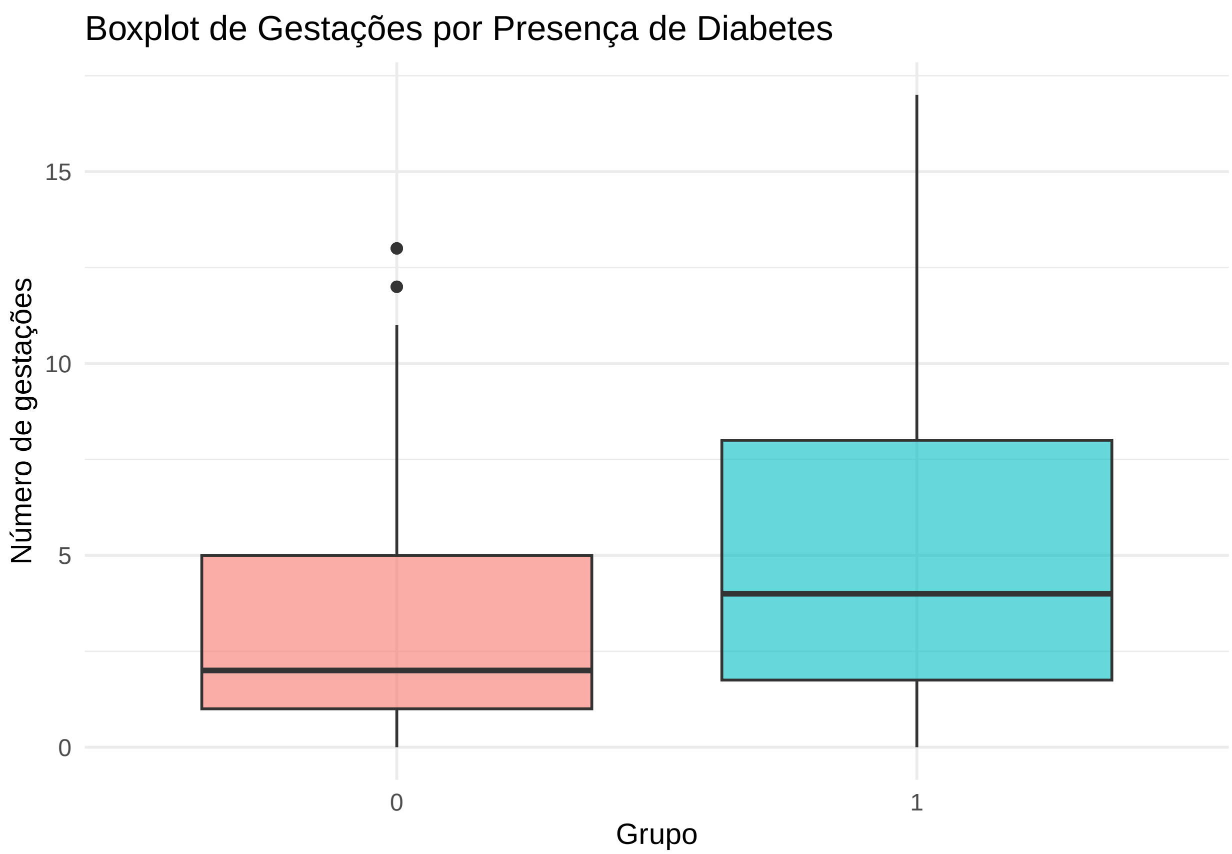
Table 2: Medidas Descritivas para Gestações

Outcome	Média	DP	Med	Min	Máx	CoefVar
0	3.298000	3.017185	2	0	13	91.48528
1	4.865672	3.741239	4	0	17	76.89050

```r
ggplot(df, aes(x = Pregnancies, fill = classe)) +
  geom_density(alpha = 0.4) +
  labs(
    title = "Densidade de Gestações por Presença de Diabetes",
    x = "Número de gestações",
    y = "Densidade",
    fill = "Grupo"
  ) +
  theme_minimal()
```




```r
ggplot(df, aes(x = Outcome, y = Pregnancies, fill = classe)) +
  geom_boxplot(alpha = 0.6) +
  labs(
    title = "Boxplot de Gestações por Presença de Diabetes",
    x = "Grupo",
    y = "Número de gestações"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

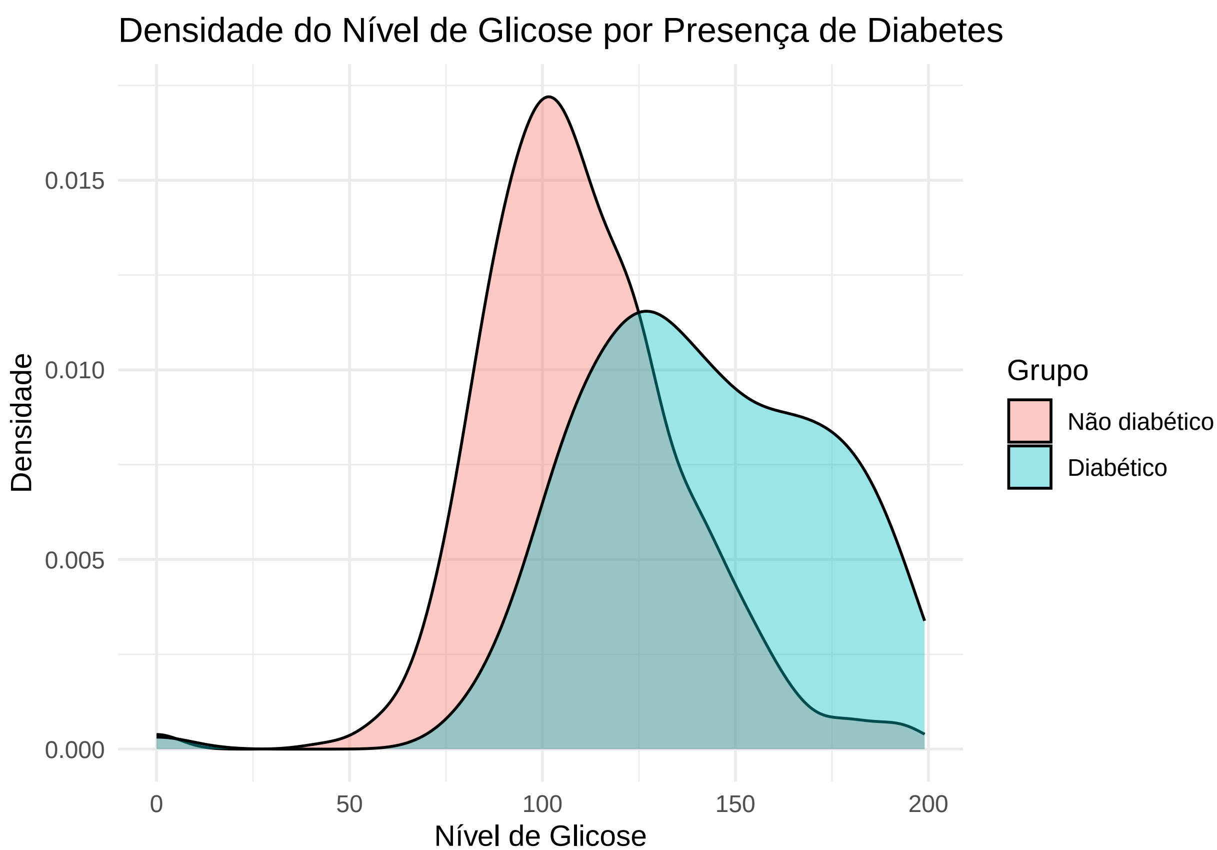


Glucose

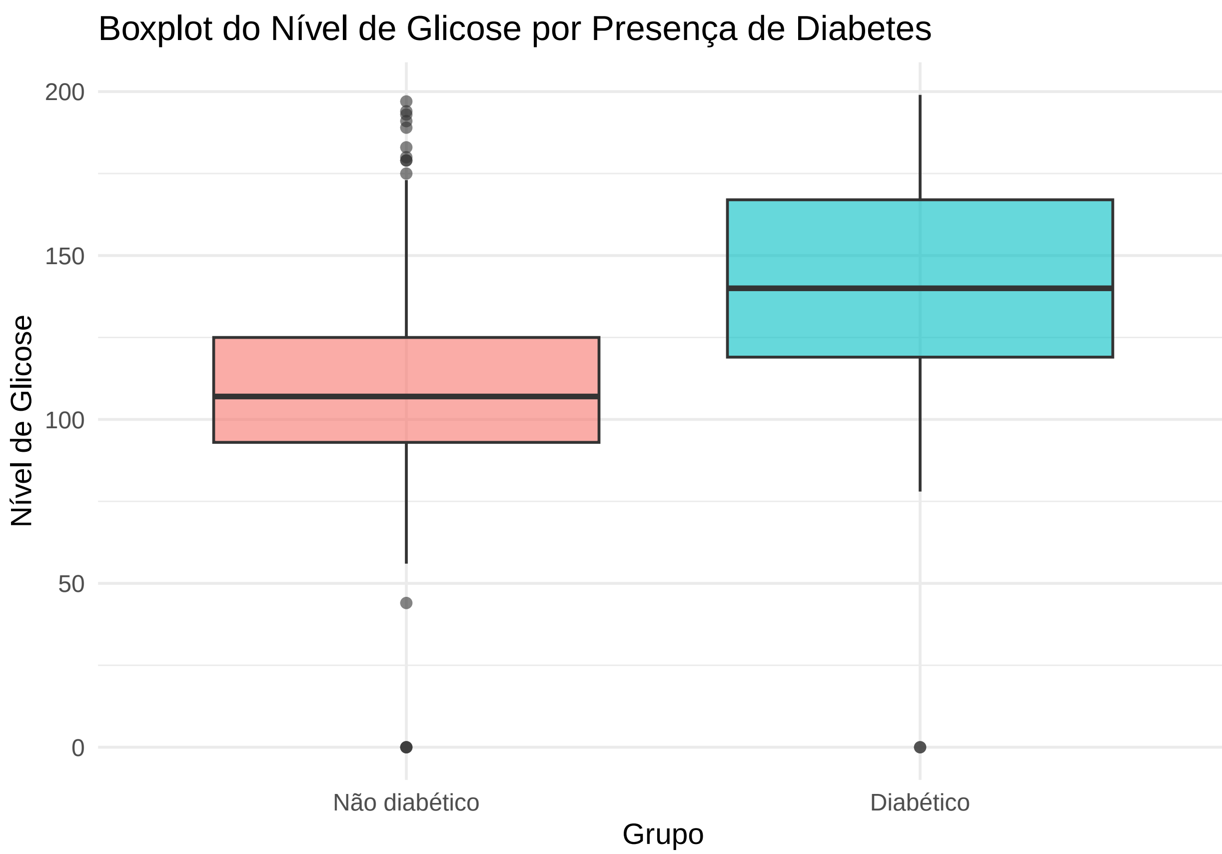
```
df %>%
  group_by(Outcome) %>%
  summarise(
    across(
      c(Glucose),
      list(
        media = ~mean(.x, na.rm = TRUE),
        dp = ~sd(.x, na.rm = TRUE),
        cv = ~sd(.x, na.rm = TRUE) / mean(.x, na.rm = TRUE) * 100,
        mediana = ~median(.x, na.rm = TRUE),
        min = ~min(.x, na.rm = TRUE),
        max = ~max(.x, na.rm = TRUE)
      ),
      .names = "{.col}_{.fn}"
    )
  )
```

```
## # A tibble: 2 x 7
##   Outcome Glucose_media Glucose_dp Glucose_cv Glucose_mediana Glucose_min
##   <fct>           <dbl>      <dbl>      <dbl>           <dbl>       <int>
## 1 0                110.       26.1       23.8             107           0
## 2 1                141.       31.9       22.6             140           0
## # i 1 more variable: Glucose_max <int>
```

```r
ggplot(df, aes(x = Glucose, fill = classe)) +
  geom_density(alpha = 0.4) +
  labs(
    title = "Densidade do Nível de Glicose por Presença de Diabetes",
    x = "Nível de Glicose",
    y = "Densidade",
    fill = "Grupo"
  ) +
  theme_minimal()
```




```r
ggplot(df, aes(x = classe, y = Glucose, fill = classe)) +
  geom_boxplot(alpha = 0.6) +
  labs(
    title = "Boxplot do Nível de Glicose por Presença de Diabetes",
    x = "Grupo",
    y = "Nível de Glicose"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

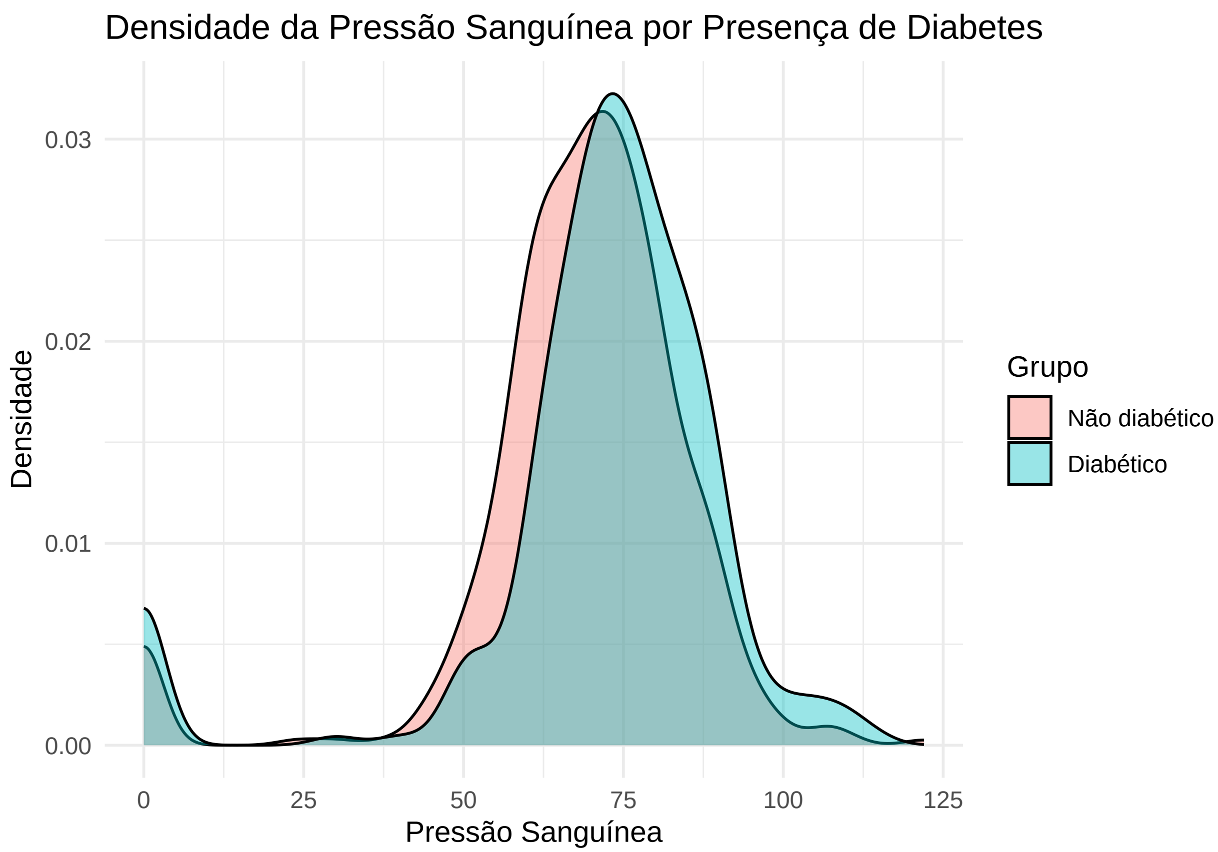


BloodPreasure

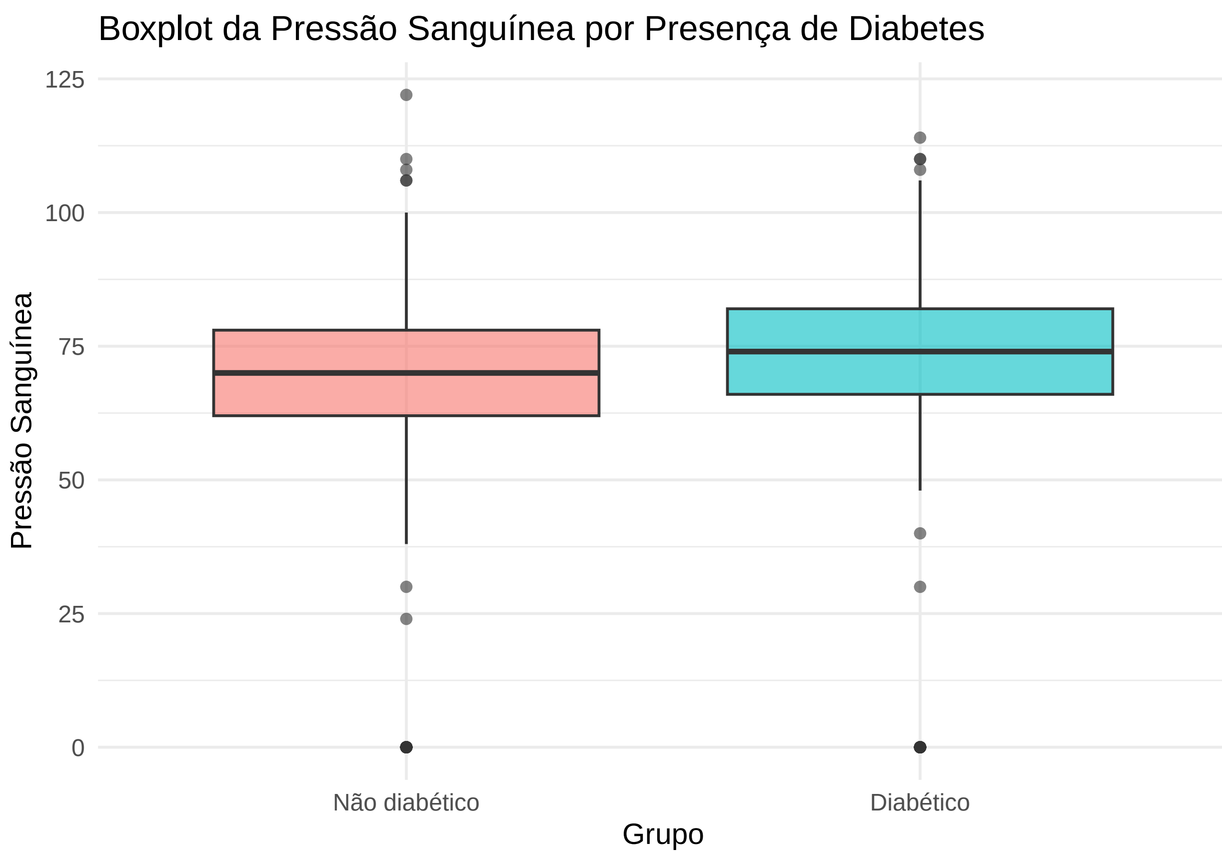
```
df %>%
  group_by(classe) %>%
  summarise(
    across(
      c(BloodPressure),
      list(
        media = ~mean(.x, na.rm = TRUE),
        dp = ~sd(.x, na.rm = TRUE),
        cv = ~sd(.x, na.rm = TRUE) / mean(.x, na.rm = TRUE) * 100,
        mediana = ~median(.x, na.rm = TRUE),
        min = ~min(.x, na.rm = TRUE),
        max = ~max(.x, na.rm = TRUE)
      ),
      .names = "{.col}_{.fn}"
    )
  )
```

```
## # A tibble: 2 x 7
##   classe                     BloodPressure_media BloodPressure_dp BloodPressure_cv
##   <fct>                                    <dbl>            <dbl>            <dbl>
## 1 "N\u00e3o diab\u00e9tico"                 68.2             18.1             26.5
## 2 "Diab\u00e9tico"                          70.8             21.5             30.3
## # i 3 more variables: BloodPressure_mediana <dbl>, BloodPressure_min <int>,
## #   BloodPressure_max <int>
```

```r
ggplot(df, aes(x = BloodPressure, fill = classe)) +
  geom_density(alpha = 0.4) +
  labs(
    title = "Densidade da Pressão Sanguínea por Presença de Diabetes",
    x = "Pressão Sanguínea",
    y = "Densidade",
    fill = "Grupo"
  ) +
  theme_minimal()
```




```r
ggplot(df, aes(x = classe, y = BloodPressure , fill = classe)) +
  geom_boxplot(alpha = 0.6) +
  labs(
    title = "Boxplot da Pressão Sanguínea por Presença de Diabetes",
    x = "Grupo",
    y = "Pressão Sanguínea"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

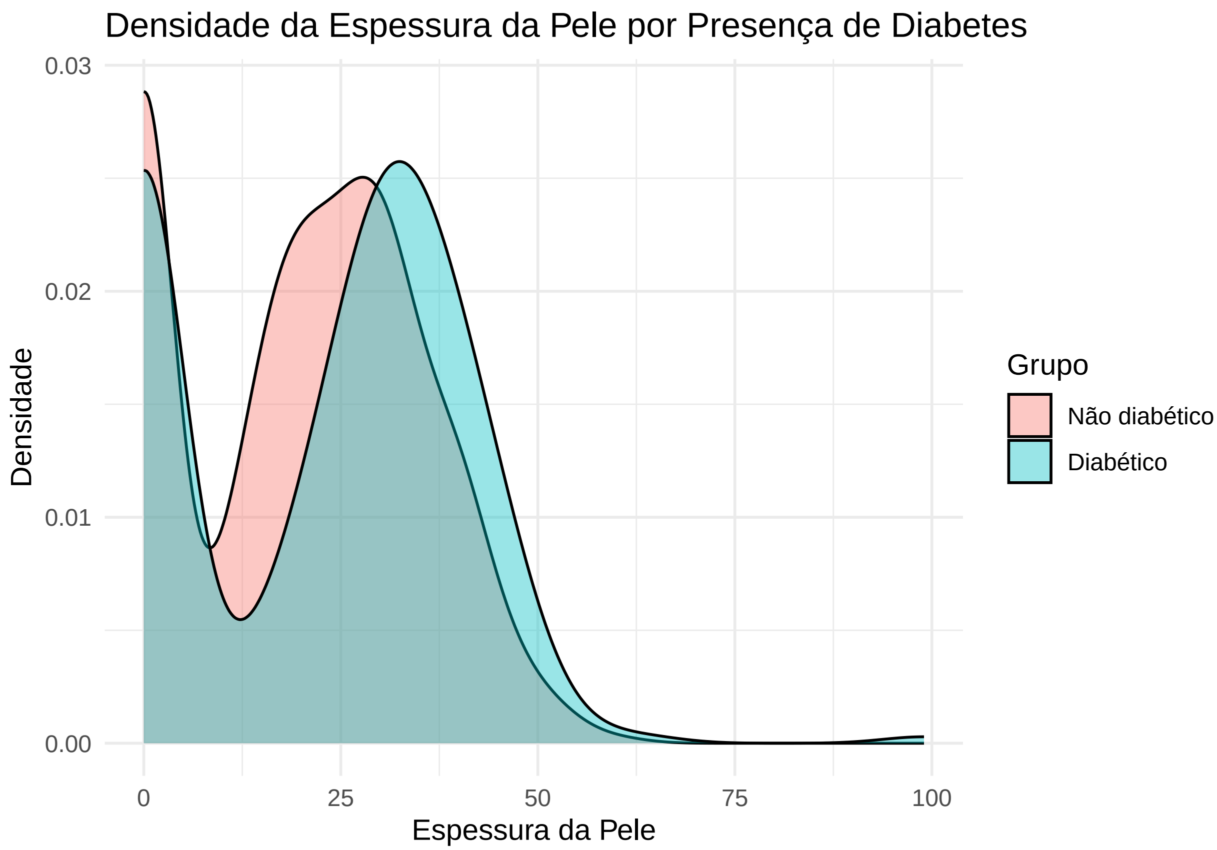


SkinThickness

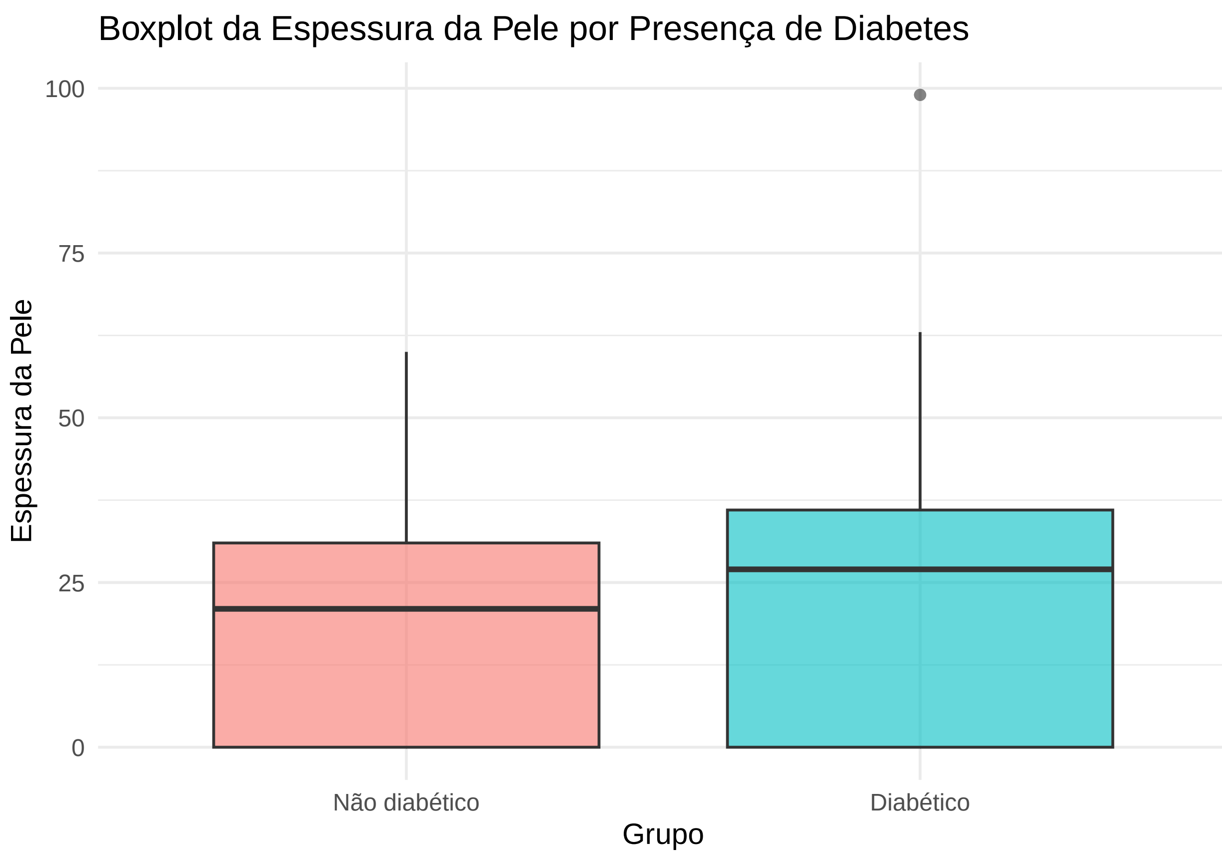
```
df %>%
  group_by(classe) %>%
  summarise(
    across(
      c(SkinThickness),
      list(
        media = ~mean(.x, na.rm = TRUE),
        dp = ~sd(.x, na.rm = TRUE),
        cv = ~sd(.x, na.rm = TRUE) / mean(.x, na.rm = TRUE) * 100,
        mediana = ~median(.x, na.rm = TRUE),
        min = ~min(.x, na.rm = TRUE),
        max = ~max(.x, na.rm = TRUE)
      ),
      .names = "{.col}_{.fn}"
    )
  )
```

```
## # A tibble: 2 x 7
##   classe                     SkinThickness_media SkinThickness_dp SkinThickness_cv
##   <fct>                                    <dbl>            <dbl>            <dbl>
## 1 "N\u00e3o diab\u00e9tico"                 19.7             14.9             75.7
## 2 "Diab\u00e9tico"                          22.2             17.7             79.8
## # i 3 more variables: SkinThickness_mediana <dbl>, SkinThickness_min <int>,
## #   SkinThickness_max <int>
```

```r
ggplot(df, aes(x = SkinThickness, fill = classe)) +
  geom_density(alpha = 0.4) +
  labs(
    title = "Densidade da Espessura da Pele por Presença de Diabetes",
    x = "Espessura da Pele",
    y = "Densidade",
    fill = "Grupo"
  ) +
  theme_minimal()
```




```r
ggplot(df, aes(x = classe, y =SkinThickness, fill = classe)) +
  geom_boxplot(alpha = 0.6) +
  labs(
    title = "Boxplot da Espessura da Pele por Presença de Diabetes",
    x = "Grupo",
    y = "Espessura da Pele"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

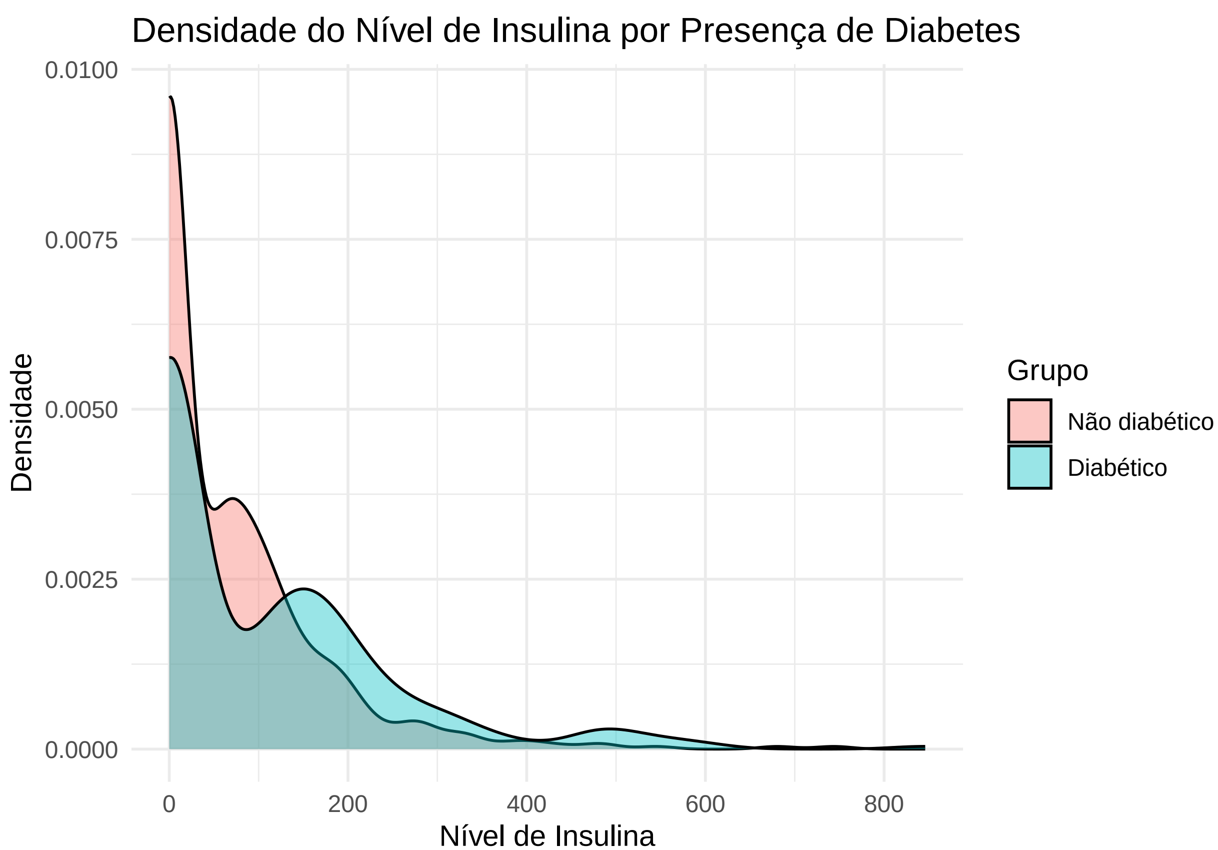


Insulin

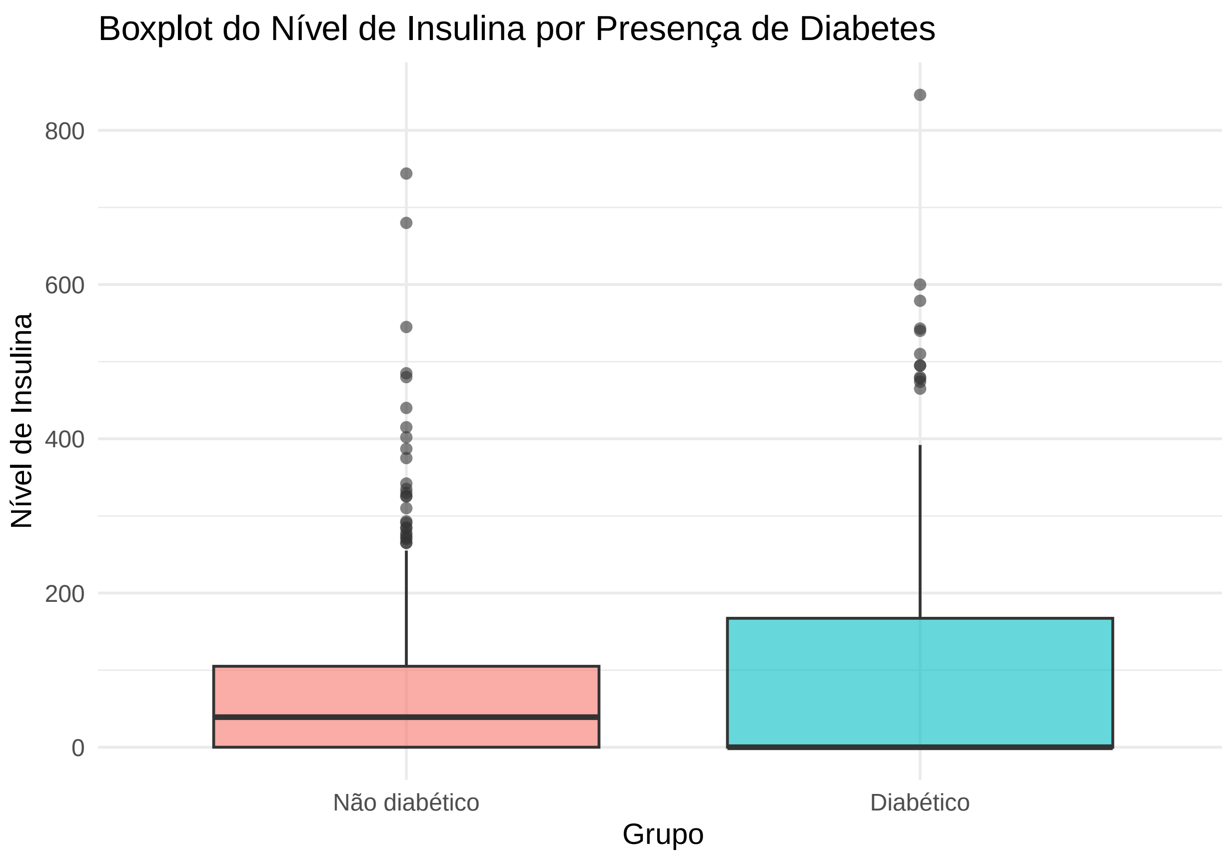
```
df %>%
  group_by(classe) %>%
  summarise(
    across(
      c(Insulin),
      list(
        media = ~mean(.x, na.rm = TRUE),
        dp = ~sd(.x, na.rm = TRUE),
        cv = ~sd(.x, na.rm = TRUE) / mean(.x, na.rm = TRUE) * 100,
        mediana = ~median(.x, na.rm = TRUE),
        min = ~min(.x, na.rm = TRUE),
        max = ~max(.x, na.rm = TRUE)
      ),
      .names = "{.col}_{.fn}"
    )
  )
```

```
## # A tibble: 2 x 7
##   classe           Insulin_media Insulin_dp Insulin_cv Insulin_mediana Insulin_min
##   <fct>                    <dbl>      <dbl>      <dbl>           <dbl>       <int>
## 1 "N\u00e3o diab\~          68.8       98.9       144.              39           0
## 2 "Diab\u00e9tico"         100.       139.        138.               0           0
## # i 1 more variable: Insulin_max <int>
```

```r
ggplot(df, aes(x = Insulin, fill = classe)) +
  geom_density(alpha = 0.4) +
  labs(
    title = "Densidade do Nível de Insulina por Presença de Diabetes",
    x = "Nível de Insulina",
    y = "Densidade",
    fill = "Grupo"
  ) +
  theme_minimal()
```




```r
ggplot(df, aes(x = classe, y = Insulin, fill = classe)) +
  geom_boxplot(alpha = 0.6) +
  labs(
    title = "Boxplot do Nível de Insulina por Presença de Diabetes",
    x = "Grupo",
    y = "Nível de Insulina"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

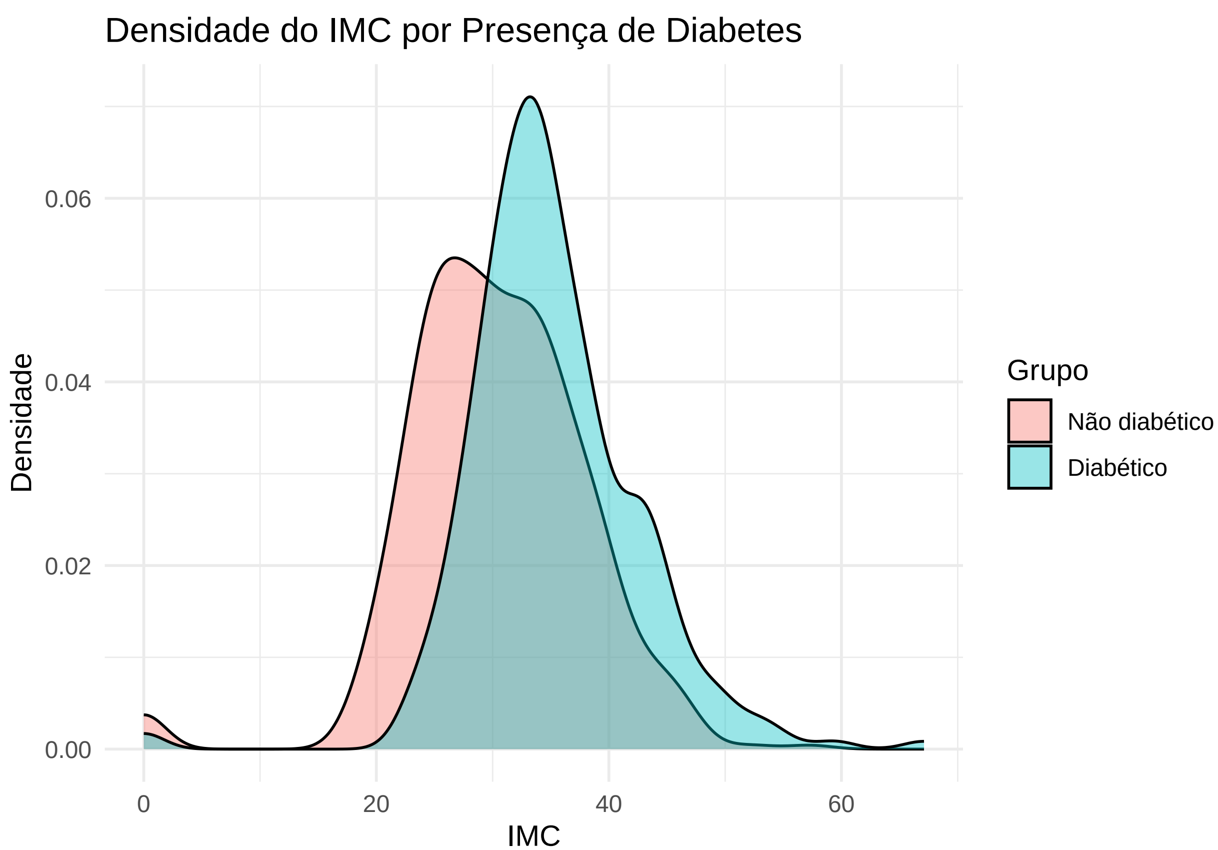


BMI

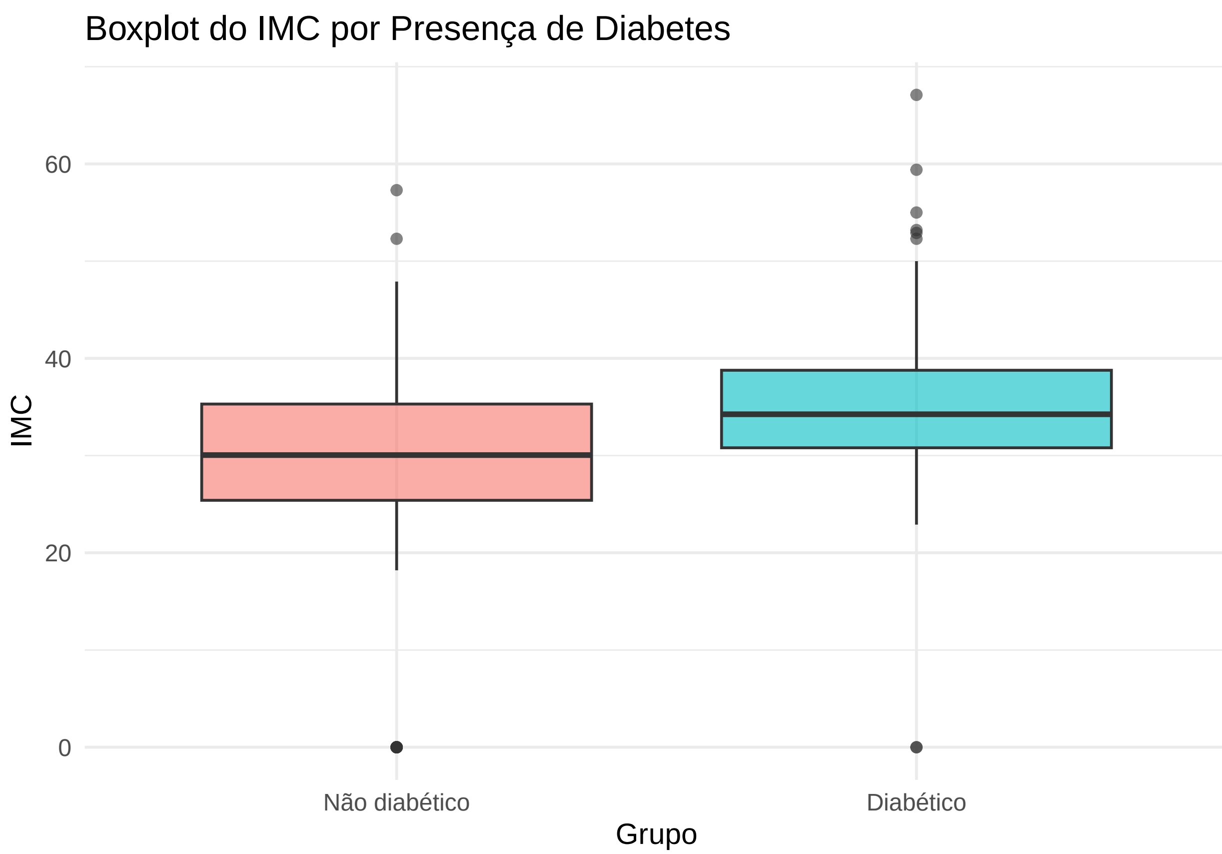
```
df %>%
  group_by(classe) %>%
  summarise(
    across(
      c(BMI),
      list(
        media = ~mean(.x, na.rm = TRUE),
        dp = ~sd(.x, na.rm = TRUE),
        cv = ~sd(.x, na.rm = TRUE) / mean(.x, na.rm = TRUE) * 100,
        mediana = ~median(.x, na.rm = TRUE),
        min = ~min(.x, na.rm = TRUE),
        max = ~max(.x, na.rm = TRUE)
      ),
      .names = "{.col}_{.fn}"
    )
  )
```

```
## # A tibble: 2 x 7
##   classe                      BMI_media BMI_dp BMI_cv BMI_mediana BMI_min BMI_max
##   <fct>                           <dbl>  <dbl>  <dbl>       <dbl>   <dbl>   <dbl>
## 1 "N\u00e3o diab\u00e9tico"        30.3   7.69   25.4        30.0       0    57.3
## 2 "Diab\u00e9tico"                 35.1   7.26   20.7        34.2       0    67.1
```

```r
ggplot(df, aes(x = BMI, fill = classe)) +
  geom_density(alpha = 0.4) +
  labs(
    title = "Densidade do IMC por Presença de Diabetes",
    x = "IMC",
    y = "Densidade",
    fill = "Grupo"
  ) +
  theme_minimal()
```




```r
ggplot(df, aes(x = classe, y = BMI, fill = classe)) +
  geom_boxplot(alpha = 0.6) +
  labs(
    title = "Boxplot do IMC por Presença de Diabetes",
    x = "Grupo",
    y = "IMC"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



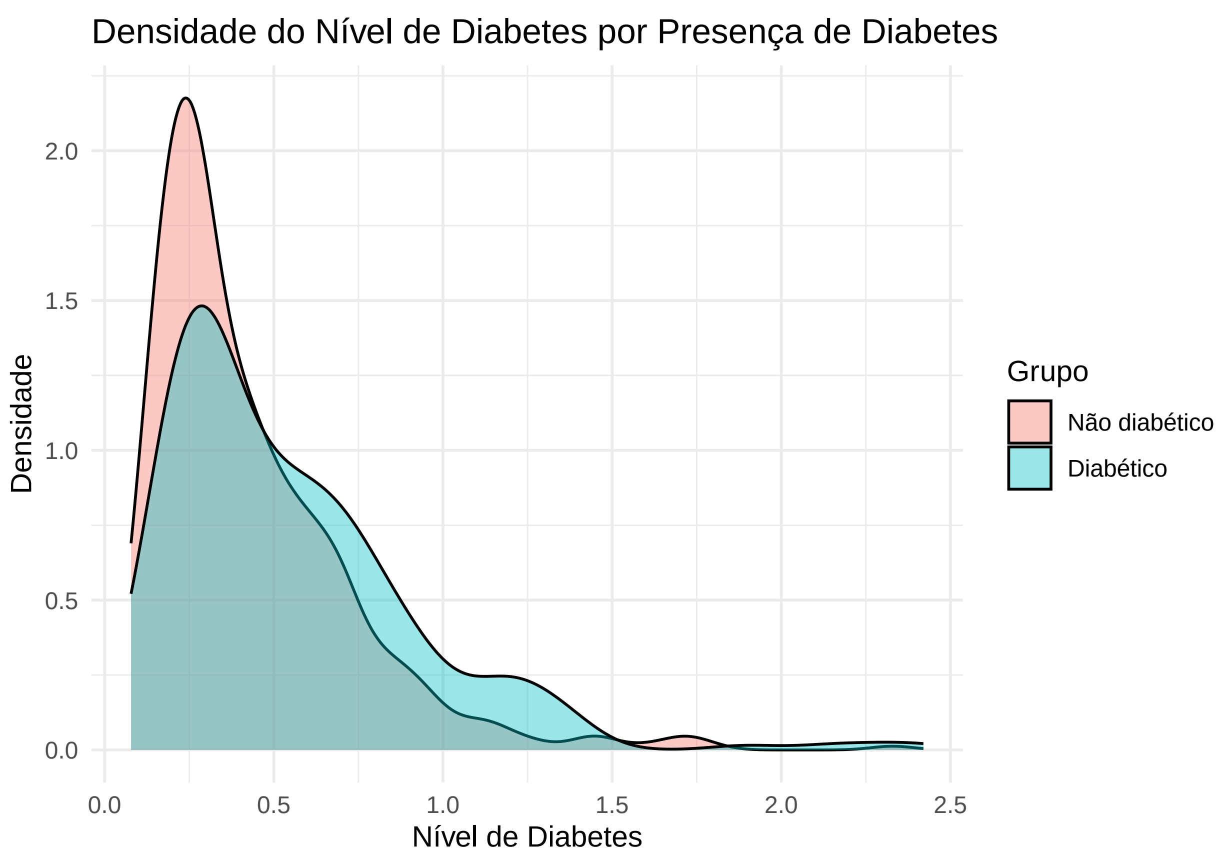
DiabetesPF

```
df %>%
  group_by(classe) %>%
  summarise(
    across(
      c(DiabetesPedigreeFunction),
      list(
        media = ~mean(.x, na.rm = TRUE),
        dp = ~sd(.x, na.rm = TRUE),
        cv = ~sd(.x, na.rm = TRUE) / mean(.x, na.rm = TRUE) * 100,
        mediana = ~median(.x, na.rm = TRUE),
        min = ~min(.x, na.rm = TRUE),
        max = ~max(.x, na.rm = TRUE)
      ),
      .names = "{.col}_{.fn}"
    )
  )
```

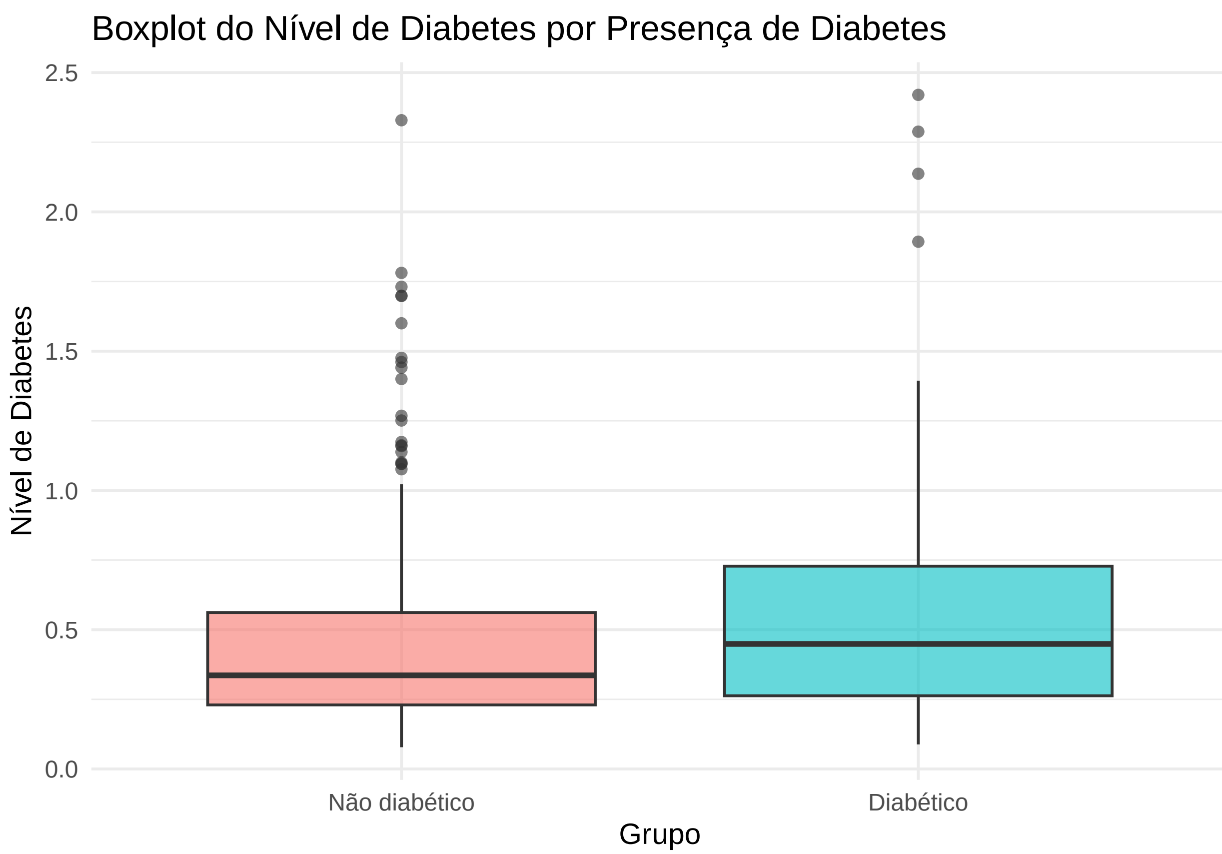
```
## # A tibble: 2 x 7
##   classe      DiabetesPedigreeFunc~1 DiabetesPedigreeFunc~2 DiabetesPedigreeFunc~3
##   <fct>                        <dbl>                  <dbl>                  <dbl>
## 1 "N\u00e3o ~                  0.430                  0.299                   69.6
## 2 "Diab\u00e~                  0.550                  0.372                   67.6
## # i abbreviated names: 1: DiabetesPedigreeFunction_media,
## #   2: DiabetesPedigreeFunction_dp, 3: DiabetesPedigreeFunction_cv
```

```
## # i 3 more variables: DiabetesPedigreeFunction_mediana <dbl>,
## #   DiabetesPedigreeFunction_min <dbl>, DiabetesPedigreeFunction_max <dbl>
```

```r
ggplot(df, aes(x = DiabetesPedigreeFunction, fill = classe)) +
  geom_density(alpha = 0.4) +
  labs(
    title = "Densidade do Nível de Diabetes por Presença de Diabetes",
    x = "Nível de Diabetes",
    y = "Densidade",
    fill = "Grupo"
  ) +
  theme_minimal()
```




```r
ggplot(df, aes(x = classe, y = DiabetesPedigreeFunction, fill = classe)) +
  geom_boxplot(alpha = 0.6) +
  labs(
    title = "Boxplot do Nível de Diabetes por Presença de Diabetes",
    x = "Grupo",
    y = "Nível de Diabetes"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

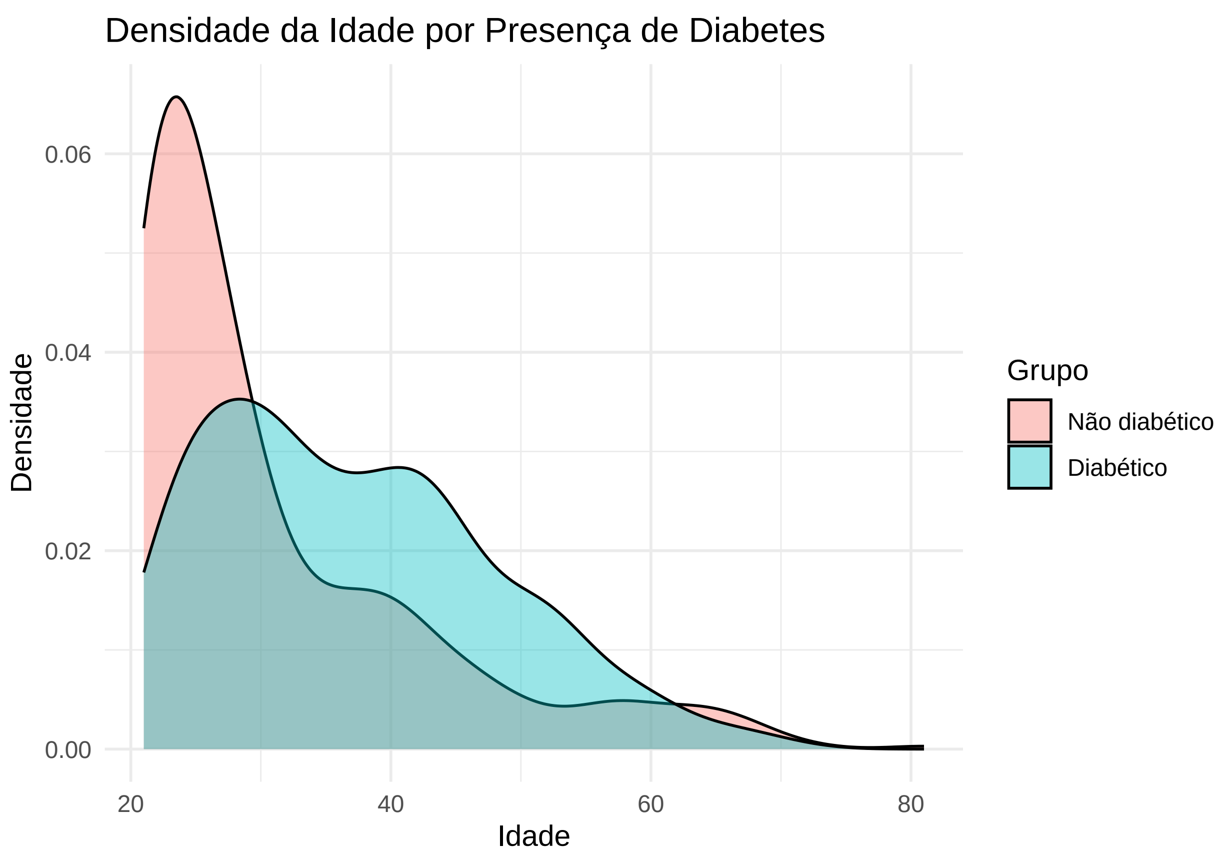


Age

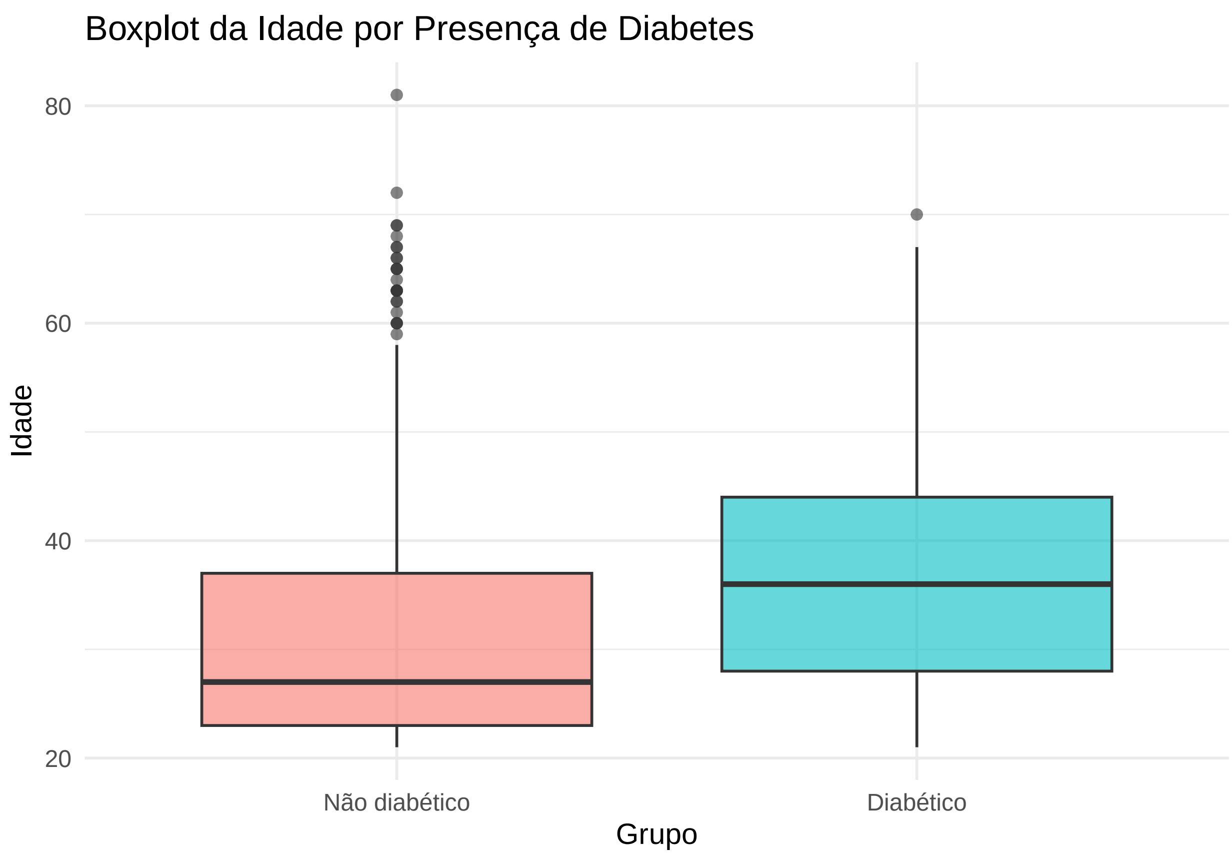
```
df %>%
  group_by(classe) %>%
  summarise(
    across(
      c(Age),
      list(
        media = ~mean(.x, na.rm = TRUE),
        dp = ~sd(.x, na.rm = TRUE),
        cv = ~sd(.x, na.rm = TRUE) / mean(.x, na.rm = TRUE) * 100,
        mediana = ~median(.x, na.rm = TRUE),
        min = ~min(.x, na.rm = TRUE),
        max = ~max(.x, na.rm = TRUE)
      ),
      .names = "{.col}_{.fn}"
    )
  )
```

```
## # A tibble: 2 x 7
##   classe                      Age_media Age_dp Age_cv Age_mediana Age_min Age_max
##   <fct>                           <dbl>  <dbl>  <dbl>       <dbl>   <int>   <int>
## 1 "N\u00e3o diab\u00e9tico"        31.2   11.7   37.4          27      21      81
## 2 "Diab\u00e9tico"                 37.1   11.0   29.6          36      21      70
```

```r
ggplot(df, aes(x = Age, fill = classe)) +
  geom_density(alpha = 0.4) +
  labs(
    title = "Densidade da Idade por Presença de Diabetes",
    x = "Idade",
    y = "Densidade",
    fill = "Grupo"
  ) +
  theme_minimal()
```




```r
ggplot(df, aes(x = classe, y = Age, fill = classe)) +
  geom_boxplot(alpha = 0.6) +
  labs(
    title = "Boxplot da Idade por Presença de Diabetes",
    x = "Grupo",
    y = "Idade"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



Modelo

```
m_null <- glm(Outcome ~ 1,
        data = df,
        family = binomial(link = "logit"))

m_full = glm(Outcome ~ Pregnancies + Glucose +
          BloodPressure + SkinThickness + Insulin + BMI +
          DiabetesPedigreeFunction + Age,
        data = df,
        family = binomial(link = "logit"))

m_final = step(m_full,
     scope = list(lower = m_null,
                  upper = m_full),
     direction = "both")
```

```
## Start:  AIC=741.45
## Outcome ~ Pregnancies + Glucose + BloodPressure + SkinThickness +
##     Insulin + BMI + DiabetesPedigreeFunction + Age
##
##                            Df Deviance    AIC
## - SkinThickness             1   723.45 739.45
## - Insulin                   1   725.19 741.19
## <none>                          723.45 741.45
## - Age                       1   725.97 741.97
```

```
## - BloodPressure              1   729.99 745.99
## - DiabetesPedigreeFunction  1   733.78 749.78
## - Pregnancies               1   738.68 754.68
## - BMI                       1   764.22 780.22
## - Glucose                   1   838.37 854.37
##
## Step:  AIC=739.45
## Outcome ~ Pregnancies + Glucose + BloodPressure + Insulin + BMI +
##     DiabetesPedigreeFunction + Age
##
##                            Df Deviance    AIC
## <none>                          723.45 739.45
## - Insulin                   1   725.46 739.46
## - Age                       1   725.97 739.97
## + SkinThickness             1   723.45 741.45
## - BloodPressure             1   730.13 744.13
## - DiabetesPedigreeFunction  1   733.92 747.92
## - Pregnancies               1   738.69 752.69
## - BMI                       1   768.77 782.77
## - Glucose                   1   840.87 854.87
```

De acordo com o gráfico abaixo, podemos notar que os resíduos estão dentro dos envelopes simulados e variando em torno da curva, indicando bom ajuste do modelo.

```r
hnp(m_final, pch = 19,
    ylab = "Resíduo Componente do Desvio",
    xlab = "HN(0,1)")
```

```
## Binomial model
```

```r
X = model.matrix(m_final)
W = diag(m_final$weights)
H = sqrt(W) %*% X %*% solve(t(X) %*% W %*% X) %*% t(X) %*% sqrt(W)
h = diag(H)
```

De acordo com o gráfico abaixo, podemos notar que os valores da medida h são bastante baixos, indicando que não há nenhuma observação realmente influente.

```r
id = 1:nrow(df)
plot(id, h, pch = 19,
     main = "Gráfico da Medida de Alavancagem",
     xlab = "Índice da Observação",
     ylab = "Medida h")
```

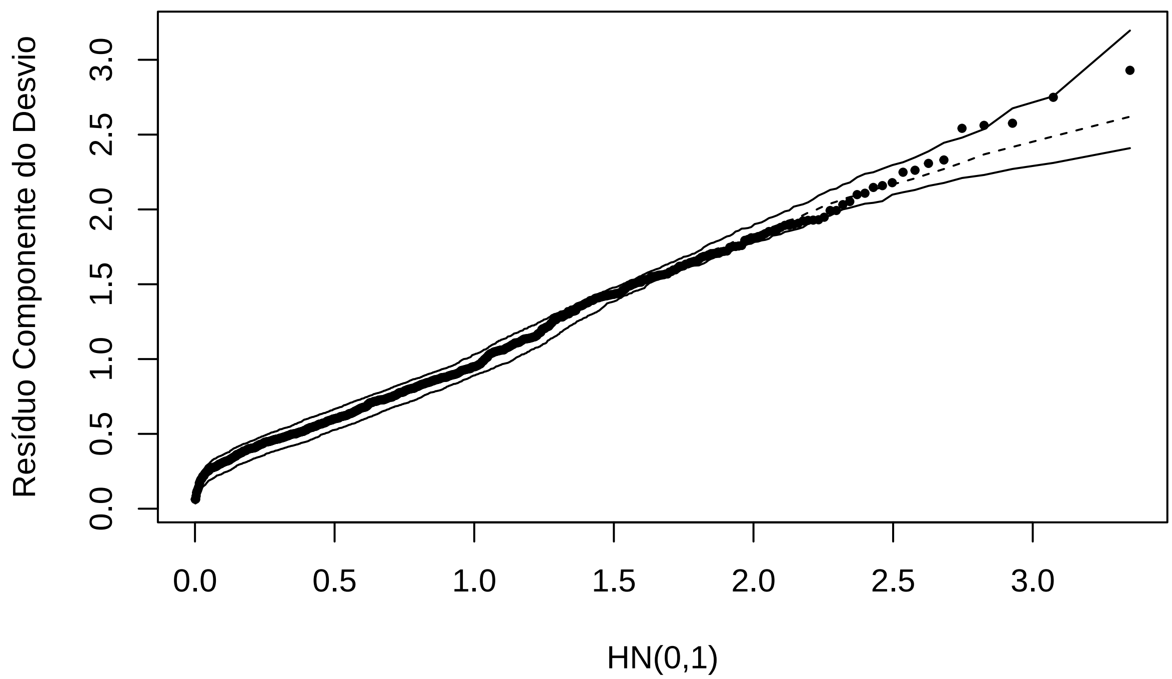
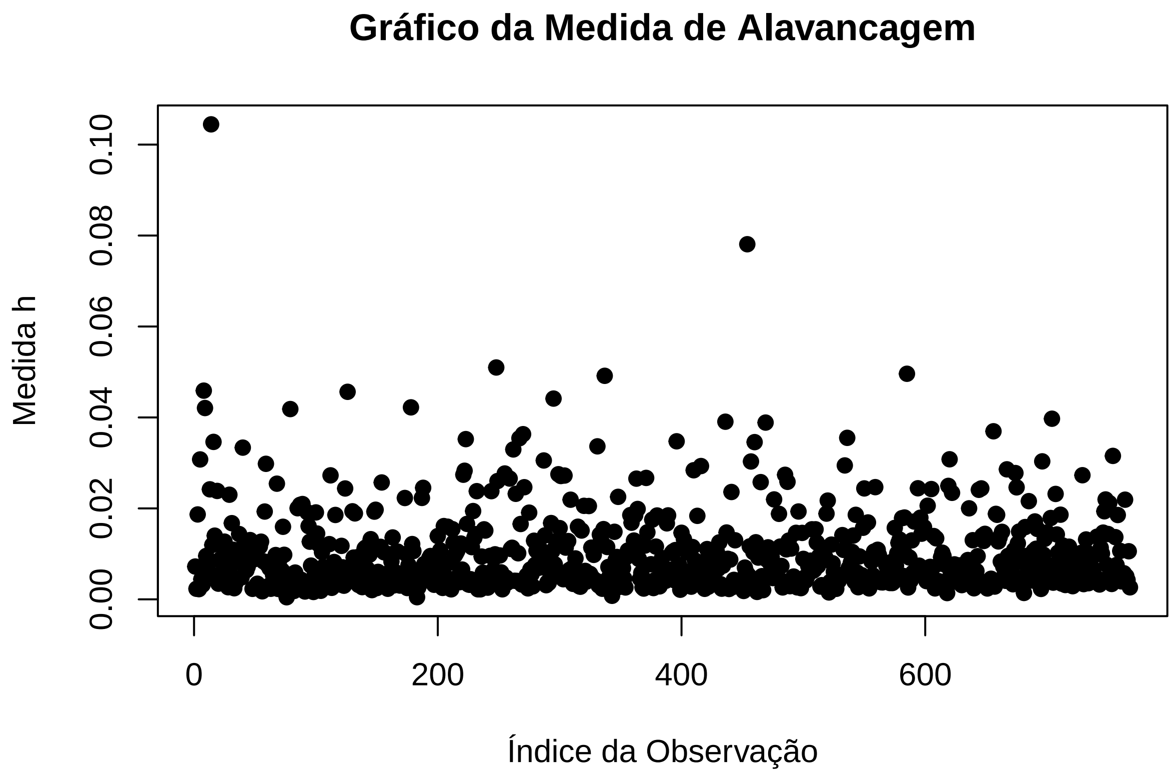


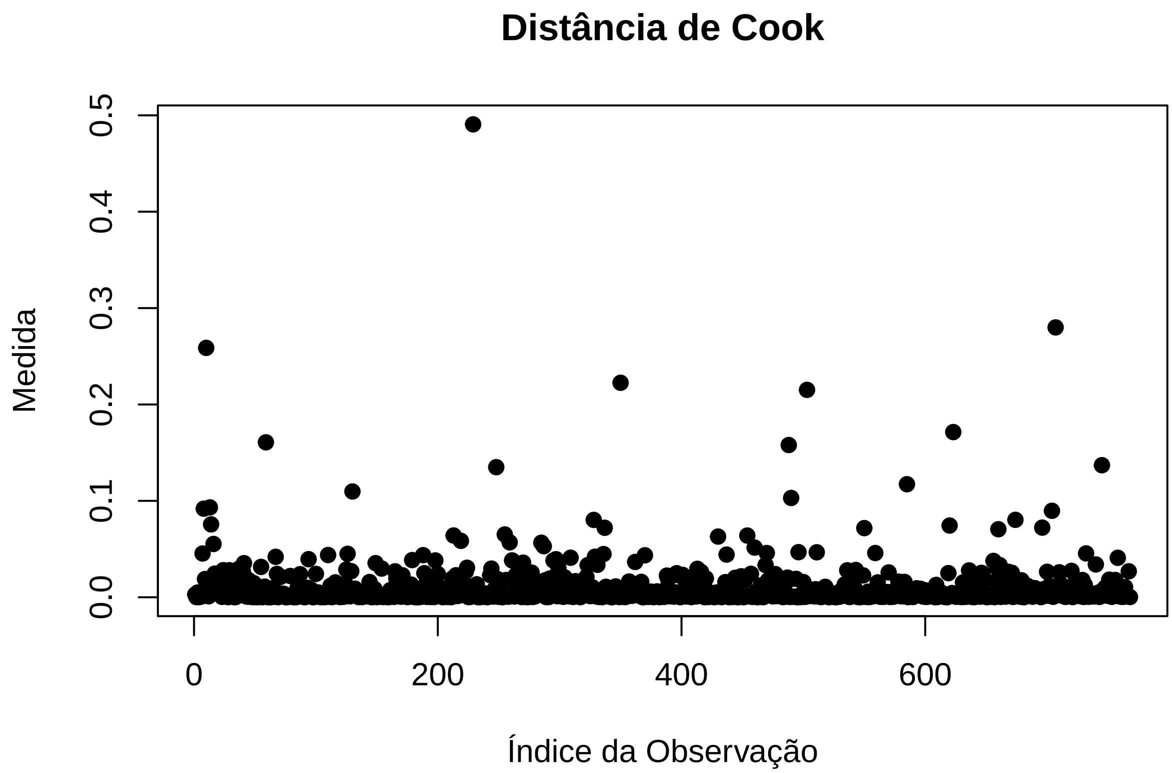
Figure 1: Gráfico de Probabilidade Meio Normal




```r
rp = resid(m_final, type= "pearson")
ts = rp/sqrt(1- h)
```

O gráfico abaixo da distância de Cook confirma o gráfico de alavancagem

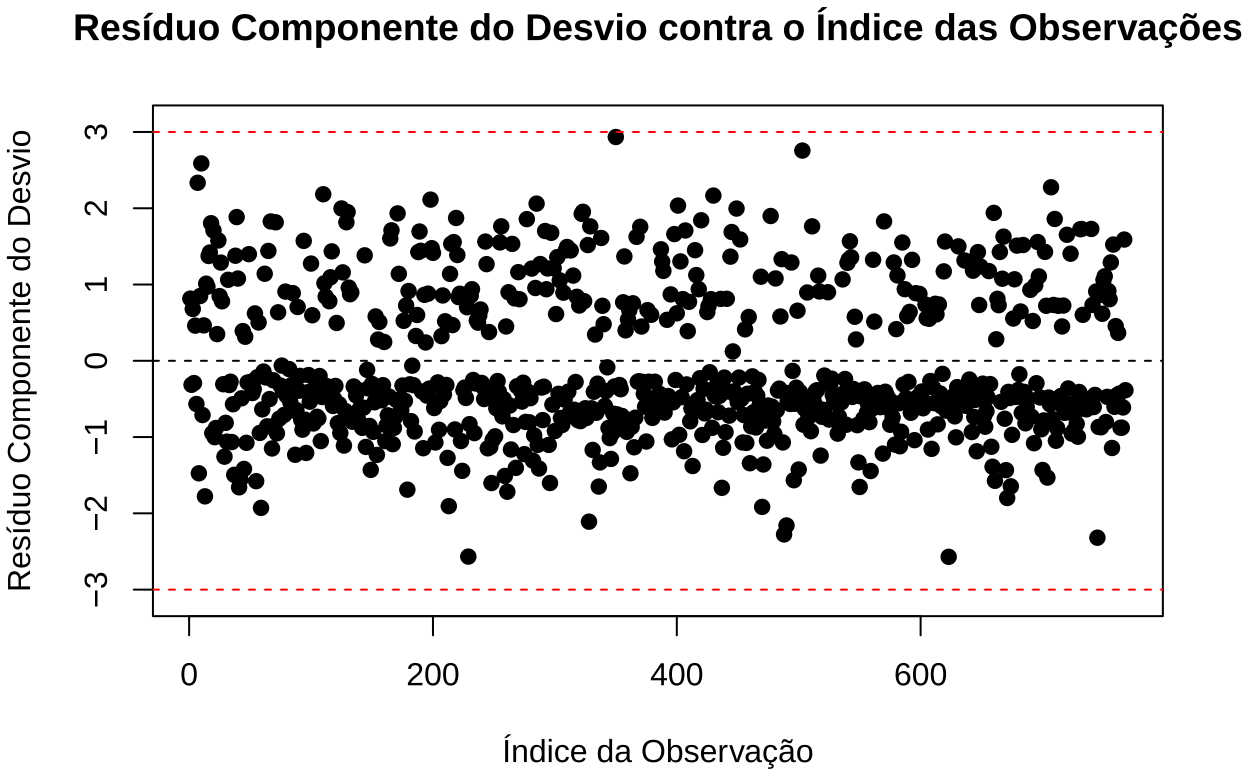
```r
cook = (ts^2) * (h/(1 - h))

id = 1:nrow(df)
plot(id, cook, pch = 19,
     main = "Distância de Cook",
     xlab = "Índice da Observação",
     ylab = "Medida")
```




```r
rd = resid(m_final, type= "deviance")
td = rd*sqrt(1/(1-h))
```

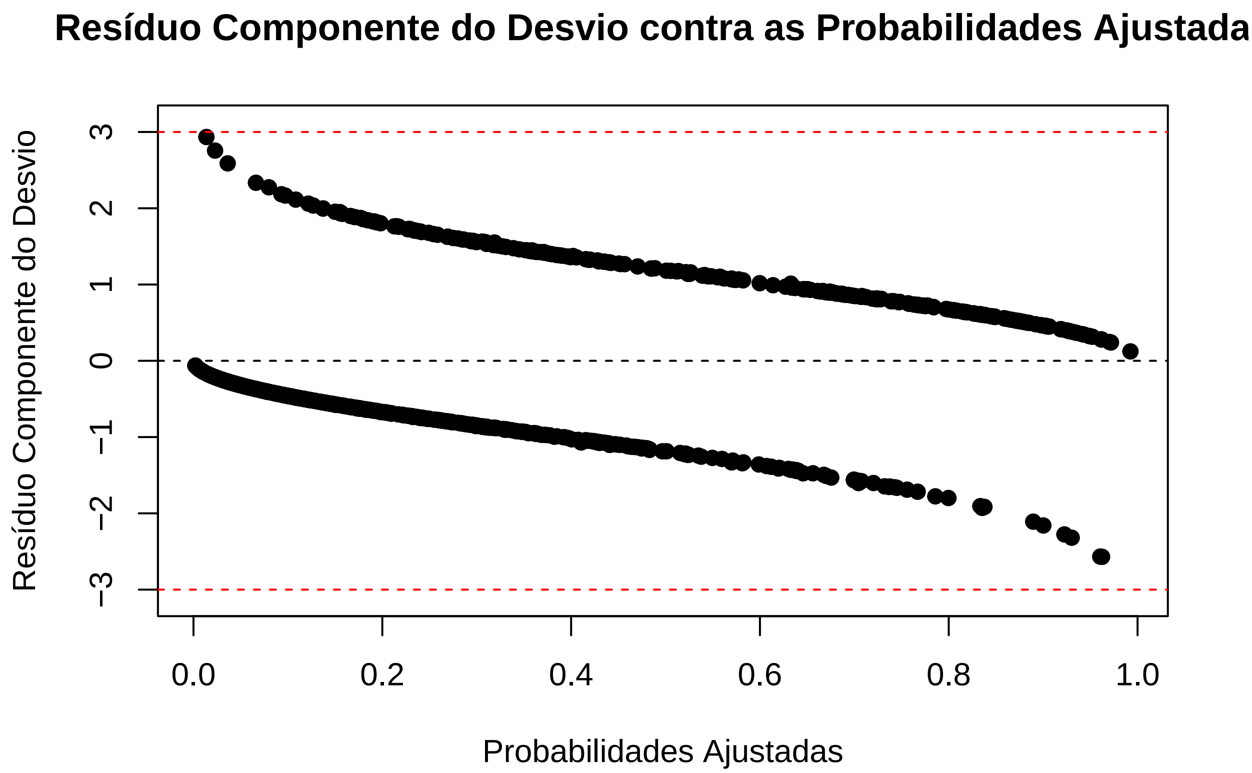
De acordo com o gráfico abaixo, notemos que os resíduos estão variando de maneira aleatório dentro das faixas, indicando bom ajuste da variabilidade.

```r
plot(id, td, pch = 19, ylim = c(-3.1, 3.1),
     main = "Resíduo Componente do Desvio contra o Índice das Observações",
     xlab = "Índice da Observação",
     ylab = "Resíduo Componente do Desvio")
abline(h = 3, col = "red", lty = 2)
abline(h = 0, lty = 2)
abline(h = -3, col = "red", lty = 2)
```



Podemos notar que, de acordo com o gráfico abaixo, temos evidências de um bom ajuste do modelo.

```
plot(m_final$fitted.values, td, pch = 19, ylim = c(-3.1, 3.1),
     main = "Resíduo Componente do Desvio contra as Probabilidades Ajustadas",
     xlab = "Probabilidades Ajustadas",
     ylab = "Resíduo Componente do Desvio")
abline(h = 3, col = "red", lty = 2)
abline(h = 0, lty = 2)
abline(h = -3, col = "red", lty = 2)
```



Inferencia

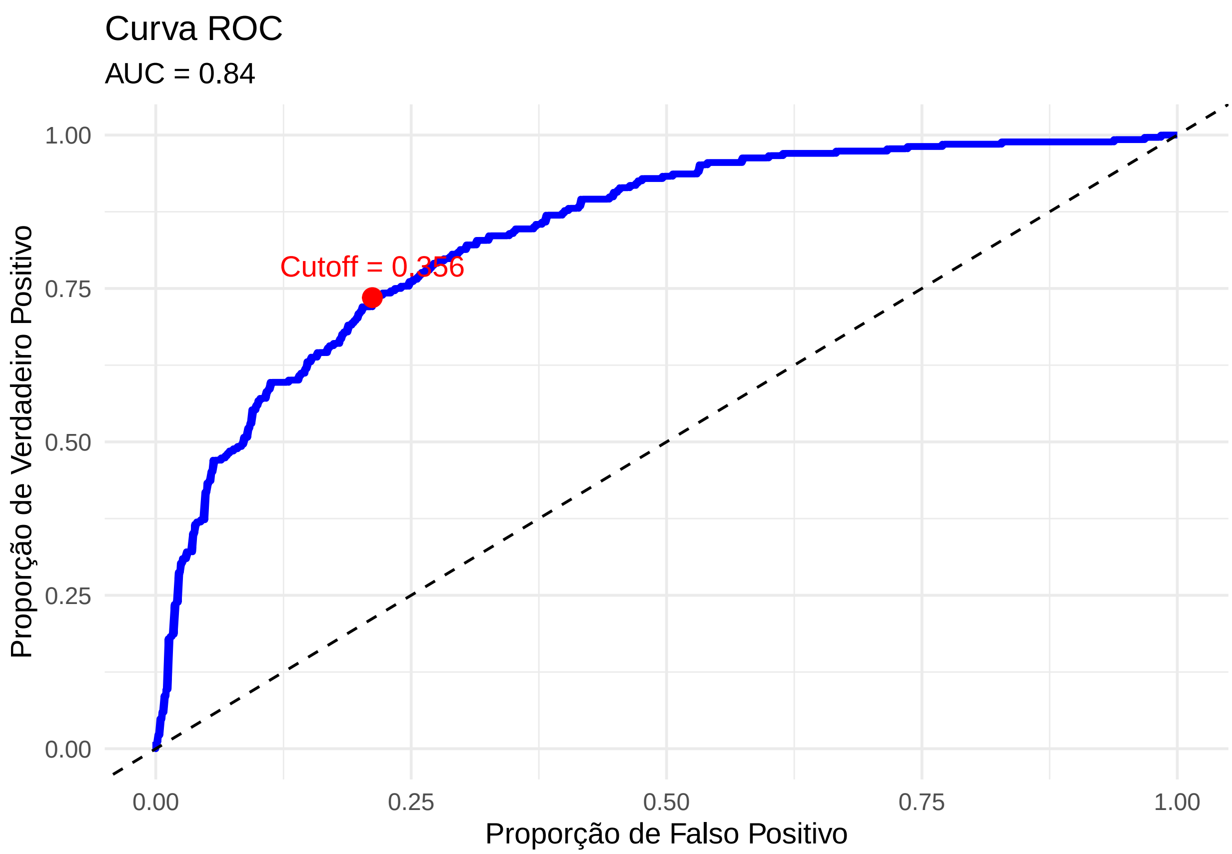
```r
summary(m_final)
```

```
##
## Call:
## glm(formula = Outcome ~ Pregnancies + Glucose + BloodPressure +
##     Insulin + BMI + DiabetesPedigreeFunction + Age, family = binomial(link = "logit"),
##     data = df)
##
## Coefficients:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)              -8.4051362  0.7167033 -11.727  < 2e-16 ***
## Pregnancies               0.1231724  0.0320688   3.841 0.000123 ***
## Glucose                   0.0351123  0.0036625   9.587  < 2e-16 ***
## BloodPressure            -0.0132136  0.0051537  -2.564 0.010350 *
## Insulin                  -0.0011570  0.0008142  -1.421 0.155275
## BMI                       0.0900886  0.0144619   6.229 4.68e-10 ***
## DiabetesPedigreeFunction  0.9475954  0.2980063   3.180 0.001474 **
## Age                       0.0147888  0.0092897   1.592 0.111393
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 993.48  on 767  degrees of freedom
```

```
## Residual deviance: 723.45  on 760  degrees of freedom
## AIC: 739.45
##
## Number of Fisher Scoring iterations: 5
```

```r
prob <- predict(m_final, type = "response")
roc_obj <- roc(df$Outcome, prob)

best_coords <- coords(roc_obj,
                      "best",
                      ret = c("threshold",
                              "sensitivity",
                              "specificity"),
                      transpose = FALSE)

cutoff <- as.numeric(best_coords["threshold"])
sens   <- as.numeric(best_coords["sensitivity"])
spec   <- as.numeric(best_coords["specificity"])

fpr <- 1 - spec
```

```r
roc_df <- data.frame(
  TPR = roc_obj$sensitivities,
  FPR = 1 - roc_obj$specificities
)

cutoff_df <- data.frame(
  FPR = fpr,
  TPR = sens
)

ggplot(roc_df, aes(x = FPR, y = TPR)) +
  geom_line(linewidth = 1.2, color = "blue") +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
  geom_point(data = cutoff_df,
             aes(x = FPR, y = TPR),
             color = "red",
             size = 3) +
  annotate("text",
           x = fpr,
           y = sens,
           label = paste0("Cutoff = ",
                          round(cutoff, 3)),
           vjust = -1,
           color = "red") +
  labs(title = "Curva ROC",
       subtitle = paste0("AUC = ",
                         round(as.numeric(auc(roc_obj)), 3)),
       x = "Proporção de Falso Positivo",
       y = "Proporção de Verdadeiro Positivo") +
  theme_minimal()
```




```r
classe_predita <- ifelse(prob >= cutoff, 1, 0)
tabela <- table(Predito = classe_predita,
                Observado = df$Outcome)
VP <- tabela["1","1"]
VN <- tabela["0","0"]
FP <- tabela["1","0"]
FN <- tabela["0","1"]

n <- sum(tabela)
ACC  <- (VP + VN) / n
SENS <- VP / (VP + FN)
ESPEC <- VN / (VN + FP)
FPR  <- 1 - ESPEC
FNR  <- 1 - SENS

resultado <- data.frame(
  Metrica = c("Acuracia",
              "Sensibilidade",
              "Especificidade",
              "Falso Positivo",
              "Falso Negativo"),
  Valor = c(ACC, SENS, ESPEC, FPR, FNR)
)

resultado
```

```
##           Metrica     Valor
```

```
## 1        Acuracia 0.7695312
## 2  Sensibilidade 0.7350746
## 3 Especificidade 0.7880000
## 4 Falso Positivo 0.2120000
## 5 Falso Negativo 0.2649254
```